

LIGHT POSTERIOR EXCHANGE FOR COMMUNICATION-EFFICIENT DISTRIBUTED LMB FUSION

Anonymous ICASSP Submission

ABSTRACT

Distributed labeled multi-Bernoulli (LMB) fusion often exchanges full Gaussian-mixture posteriors even when the receiver-side fusion step ultimately consumes label-wise moment-matched Bernoulli statistics. This paper studies a simpler communication lever: keep the effective Kullback-Leibler average (KLA) fusion graph intact, but replace full posterior messages with light posteriors containing per-label existence probabilities and single-Gaussian moments. On a 4+4 multisensor formation benchmark, a periodic light-posterior exchange matches the full-posterior baseline in tracking and consensus metrics over 50 held-out trials while reducing estimated communication by 58.6%. The result suggests that, for single-round GA/KLA-LMB fusion, payload compression can be safer than event-triggered graph sparsification.

Index Terms— Distributed multi-object tracking, labeled random finite sets, LMB filter, KLA fusion, communication-efficient sensing

1. INTRODUCTION

Distributed multi-object tracking lets a sensor network combine complementary views without centralizing all measurements. Labeled random finite-set filters, especially labeled multi-Bernoulli (LMB) filters, are well suited to this task because they represent target existence, state uncertainty, and identity in one posterior [1, 2, 3]. Their communication cost can be high: each transmitted LMB posterior may contain many labels and multiple Gaussian components per label.

Most communication-saving designs reduce how often nodes communicate or which neighbors they contact. That can be risky for conservative fusion rules such as covariance intersection and Kullback-Leibler average (KLA) fusion [4, 5]. The relevant object is not only the nominal graph, but the effective graph of posterior messages that actually enter the fusion step. Sparsifying this graph can reduce bytes while degrading consensus and label agreement.

This paper instead keeps the effective graph intact and compresses the payload. The observation is sim-

ple: in our single-round GA/KLA-LMB fusion path, incoming components are aligned by label and reduced to Bernoulli moments before fusion. When the receiver consumes those moments, transmitting every mixture component can be unnecessary. We therefore send a light posterior containing each active label’s existence probability, moment-matched mean, and moment-matched covariance over the same static communication graph used by the full-posterior baseline.

Over 50 held-out trials on an eight-sensor 4+4 formation benchmark, this periodic light-posterior exchange reduces estimated communication by 58.6% while matching full-posterior exchange in local E-OSPA, consensus OSPA, position disagreement, cardinality dispersion, and effective-weight graph connectivity. Dynamic-topology variants are less stable, supporting the paper’s central design rule: preserve information paths first, then compress posterior payloads.

Our contributions are a graph-preserving light posterior exchange protocol for distributed LMB fusion, an effective-fusion-graph perspective that separates payload compression from message-path sparsification, and a held-out 50-trial validation showing full-posterior-equivalent behavior with about 58% lower estimated communication.

2. RELATED WORK

RFS tracking. Random finite-set filters provide a Bayesian framework for tracking uncertain target cardinality and measurement association. PHD filters introduced a tractable first-moment recursion [6]; labeled RFS models add persistent identities [1]; and LMB filters provide a practical independent-Bernoulli approximation [2]. Multi-sensor GLMB/LMB variants extend this framework to cooperative sensing [3]. We do not modify the LMB recursion; we study how much of the posterior must be communicated before distributed fusion.

Conservative fusion. When cross-correlations are unknown, conservative rules avoid overconfident fusion. Covariance intersection handles Gaussian estimates [4], while KLA/GCI extends the log-opinion-pool idea to probability densities [5]. Related log-opinion-pool analyses appear in [7]. RFS variants include exponential-

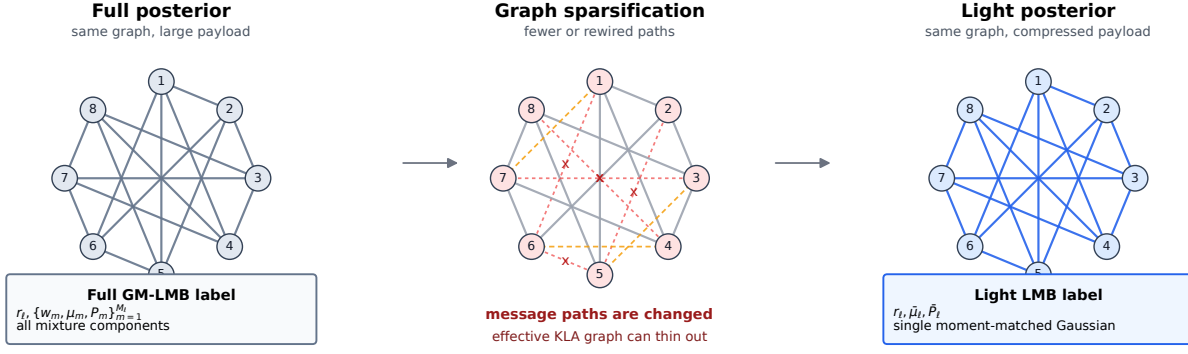


Fig. 1. Two ways to reduce communication in distributed LMB fusion. Graph sparsification changes which posterior messages enter the KLA update, while light posterior exchange keeps the same effective graph and compresses each label payload from a Gaussian mixture to moment-matched Bernoulli statistics.

mixture PHD fusion [8], multi-Bernoulli GCI fusion [9], and robust distributed labeled RFS fusion [10, 11]. These works define how received densities should be fused; our question is what representation needs to be transmitted when the receiver already reduces labels to moments.

Communication-aware fusion. Prior work handles limited links through selective transmission, adaptive scheduling, consensus, and reliability-aware fusion. Examples include limited-communication labeled RFS fusion [12], consensus LMB filtering [13], reliability-weighted distributed RFS fusion [14], geometry-aware fusion [15], and average-density fusion approximations [16]. These approaches often change message timing, neighbor choice, or fusion weights. Our work separates that graph-level problem from payload compression: we keep the message paths fixed and ask whether a lighter label representation is sufficient for the KLA update.

3. LIGHT POSTERIOR EXCHANGE

3.1. Distributed LMB Fusion Baseline

At time k , sensor i maintains an LMB posterior

$$\pi_i = \{(r_i^\ell, p_i^\ell(x)) : \ell \in \mathcal{L}_i\}, \quad (1)$$

where r_i^ℓ is the existence probability of label ℓ and $p_i^\ell(x)$ is represented by a Gaussian mixture. In the full-posterior baseline, each node periodically sends its complete GM-LMB posterior to every neighbor in a fixed communication graph. After packet drops are realized, each receiver aligns common labels and applies a single synchronous KLA-style fusion step with the same fusion weights for all compared variants.

For a receiver i and the delivered neighbor set $\mathcal{N}_i(k)$,

the fusion step can be written abstractly as

$$\bar{\pi}_i \propto \pi_i^{\omega_{ii}} \prod_{j \in \mathcal{N}_i(k)} \pi_j^{\omega_{ij}}, \quad \sum_{j \in \mathcal{N}_i(k) \cup \{i\}} \omega_{ij} = 1. \quad (2)$$

The implementation aligns labels and moment-matches components inside the LMB fusion path. Thus full Gaussian-mixture transmission can be redundant: the receiver does not preserve arbitrary mixture structure after the single-round fusion update.

3.2. Light Posterior Message

The proposed message keeps the Bernoulli label set and existence probabilities but replaces each transmitted mixture density with a single moment-matched Gaussian. For a label density

$$p_i^\ell(x) = \sum_{m=1}^{M_i^\ell} w_{im}^\ell \mathcal{N}(x; m_{im}^\ell, P_{im}^\ell), \quad (3)$$

the light posterior sends

$$\tilde{p}_i^\ell(x) = \mathcal{N}(x; \bar{m}_i^\ell, \bar{P}_i^\ell), \quad (4)$$

with

$$\bar{m}_i^\ell = \sum_m w_{im}^\ell m_{im}^\ell, \quad (5)$$

$$\bar{P}_i^\ell = \sum_m w_{im}^\ell (P_{im}^\ell + (m_{im}^\ell - \bar{m}_i^\ell)(m_{im}^\ell - \bar{m}_i^\ell)^\top). \quad (6)$$

Thus a light message contains the label, existence probability, single mean, single covariance, and minimal book-keeping metadata. It does not change the local prediction, measurement update, label pruning, fusion weights, packet drop model, or neighbor schedule. Only the transmitted posterior representation changes.

3.3. Graph Preservation and Diagnostics

The communication policy is intentionally conservative. The main method sends light posteriors periodically over the same static topology used by the full-posterior baseline. This isolates payload compression from graph sparsification: if performance changes, the cause is posterior summarization, not missing message paths.

To verify this separation, we report effective-fusion-graph diagnostics in addition to tracking metrics. The attempted graph records scheduled neighbor messages, the delivered graph records successful transmissions, and the effective-weight graph records nonzero neighbor mass that actually enters the KLA update. We summarize each graph by the algebraic connectivity of its symmetrized adjacency or weight matrix. We also track label-level stale age and the total fusion weight assigned to self, light, and full posterior messages. The target behavior for a safe compression method is therefore not only low bytes, but also an effective-weight graph close to the full-posterior baseline.

4. EXPERIMENTS

4.1. Benchmark and Metrics

We evaluate on an eight-sensor 4+4 formation benchmark with 100 simulation steps per trial. Each node runs the same local LMB prediction and measurement update, then performs one synchronous distributed fusion round. The static communication graph has 16 undirected edges. Packet delivery is stochastic, with average node-level drop probabilities near 0.2 in the held-out run. All arms use the same seeds, measurements, target trajectories, and packet-drop draws.

We report estimated communication bytes, local tracking accuracy, and network agreement. Local accuracy is measured by the mean local E-OSPA score, while network agreement is measured by consensus OSPA, position disagreement, and cardinality dispersion [17]. Communication is estimated from the number of transmitted scalar posterior values. The trial-level pass gate requires at least 30% byte reduction relative to full posterior exchange, no more than 5% local E-OSPA degradation, and no more than 10% degradation in consensus OSPA, position disagreement, and cardinality dispersion.

4.2. Compared Arms

The primary baseline is periodic full-posterior exchange on the static graph. The proposed arm is periodic light-posterior exchange on the same graph. We also include two diagnostic arms. First, full-posterior exchange with

Table 1. Held-out 50-trial results over seeds 32–81. Lower bytes and OSPA are better.

Arm	Bytes (M)	Red.	Local	Cons.	λ_2^{eff}	Pass
Full-static	27.77	0.0%	2.065	1.950	0.373	ref.
Full-dynamic	29.47	-6.1%	2.151	2.167	0.301	0/50
Light-static	11.50	58.6%	2.065	1.950	0.373	50/50
Light-dynamic	11.57	58.3%	2.067	1.964	0.371	35/50

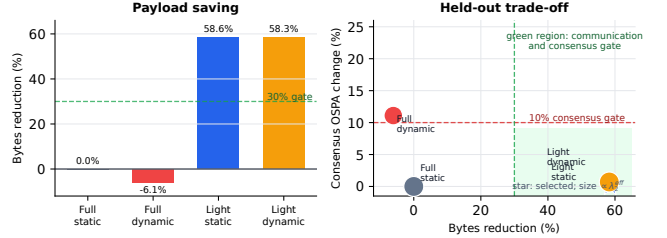


Fig. 2. Held-out aggregate trade-off. Bars show byte reduction relative to full-static exchange. The scatter plot compares byte reduction and consensus degradation; the shaded region satisfies the communication and consensus gates. Marker size is proportional to effective-weight graph connectivity.

guarded dynamic topology tests whether changing neighbors alone improves the trade-off. Second, light-posterior exchange with guarded dynamic topology tests whether the light payload remains stable when the graph is allowed to adapt. The guarded dynamic topology uses the same 16-edge budget as the static graph and falls back when connectivity checks fail; it is included as an ablation rather than as the main method.

4.3. Held-out Results

Table 1 and Fig. 2 show that the graph-preserving light posterior is the only non-reference arm that exactly matches the full-posterior baseline on both local tracking and consensus metrics. It reduces estimated communication from 27.77M to 11.50M bytes, a 58.6% reduction, while preserving the effective-weight algebraic connectivity at 0.373. Its paired trial pass count is 50/50, and its worst-case byte reduction remains above 55%.

The dynamic-topology arms clarify why the main method keeps the static graph. Dynamic topology with full posterior exchange sends slightly more bytes than the static baseline and degrades consensus OSPA by 11.1% on average; its effective-weight connectivity drops from 0.373 to 0.301. This indicates that nominally adaptive connectivity does not guarantee a better KLA information flow. Adding light payloads to guarded dynamic topology recovers most average performance, but the pass count falls to 35/50 because some trials still incur

larger agreement or cardinality deviations. Thus the robust result is not “dynamic topology plus light messages,” but the simpler combination of a fixed effective graph with light posterior payloads.

These results support the paper’s central claim. In this single-round GA/KLA-LMB setting, the dominant communication saving comes from removing unneeded mixture-detail payload, not from dropping message paths. When the receiver ultimately fuses label-wise Bernoulli moments, periodic light posterior exchange can reproduce full posterior behavior while substantially reducing communication.

5. CONCLUSION

This paper studied a narrow but practical communication question for distributed LMB fusion: whether full Gaussian-mixture posteriors must be exchanged when the receiver-side KLA update uses label-wise moment-matched Bernoulli statistics. In the validated 4+4 benchmark, preserving the static effective fusion graph and sending light posterior messages matches full posterior exchange over 50 held-out trials while reducing bytes by 58.6%. The result suggests a conservative design rule for this class of trackers: preserve effective information paths first, then compress posterior payloads. Future work should identify regimes where multimodal label densities, stronger association ambiguity, or multi-round consensus require richer messages again.

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